

MARK YANG

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EDUCATION

Northeastern University

Boston, MA (April 2028)

- BS in Electrical and Computer Engineering (3.964 GPA)
- Relevant coursework: Embedded Design, Circuits and Signals, Fundamentals of Networks, Neuromechanical Simulation of Human Movement, Differential Equations, Physics 2
- Activities: Putnam Club

Queens University Belfast

Belfast, Northern Ireland (September–December 2024)

- N.U.*in* Study Abroad Semester

Rutgers University New Brunswick

New Brunswick, NJ (June–August 2023)

- Coursework: Discrete Structures (A), Multivariable Calculus (A)

PROFESSIONAL EXPERIENCE

Northeastern University

Undergraduate Research Assistant @ NEUMove

Boston, MA (August 2025–Present)

- Built and modeled 10 different types of lower limb exoskeleton aids and simulated using mechanistic training with PyTorch and MyoAssist/MuJoCo physics environment.
- Studied different types of simulation environments (Matlab/simulink, myoassist, opensim, etc.) and how to optimize controller training across broad spectrum models.

Northeastern University

Engineering Center Aide

Boston, MA (August 2025–Present)

- Assisted the freshmen engineering class of 900+ students in Raspberry Pi Pico, Python, MATLAB, AutoCAD, and Solidworks.
- Maintain the first-year engineering makerspace and host bi-weekly workshops for students on soldering.

Synbio Technologies

Software Engineering Intern

South Brunswick, NJ (June–August 2024)

- Developed the backend JavaScript of the company webpage to track user interaction and plot which links and intractable objects had the most traffic.
- Organized a team for a complete overhaul of the main website visual design, helping change and modernize 100+ different pages.

Kumon

Assistant Teacher

Princeton, NJ (April 2023–June 2024)

Princeton Day School

Assistant Teacher

Princeton, NJ (September 2023–May 2024)

- Volunteered as an assistant teacher for the 7th grade computer science and web-development class.
- Helped build and develop the school curriculum to introduce high-level text based programming such as JavaScript and Python.
- Organized workshops and small speaker events with the head instructor to help facilitate more interest among students.

Storming Robots

Teacher/Tutor

Branchburg, NJ (May 2023–February 2024)

- Guided students in groups of 10-15 through the K.N.King C Programming textbook, creating curated problems designed towards each individual student's needs.
- Introduced middle school students to the LEGO Mindstorms robotics platform, tackling challenges in the RoboCup Junior (RCJ) competition league.

PROJECTS

Hexapode Robot

Class Project (June–July 2025)

- Worked on controlling a Hexapode Robot using the DE1-SoC intel FPGA board, designing a Quartus truth gate circuit for PWM servo control. Programmed a C++ script for the ARM System on a Chip for MMIO servo control, allowing the robot to walk, turn, dance, etc.

RoboCup Junior Maze Simulated Robot

Group Project (December 2021 – July 2022)

- Helped program the Breadth First Search algorithm for a maze traversal robot on the Erebus simulation platform. Also created all maze worlds for debug and testing. Placed 1st in the 2022 RCJ international competition hosted in Thailand.

SKILLS AND HONORS

Languages/Tools Python, Java, C, C++, MATLAB, SolidWorks, HTML5/CSS3/JS, Git, L^AT_EX, Verilog, Excel

Awards/Honors USACO Silver Division, 1st in Simulation Division 2022 RCJ Thailand